

American Magnetics, Inc.

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MAGNET SYSTEM SPECIFICATIONS

AMI JOB #021541-1-1 MAGNET #14432

TYPE: **CRYOGEN-FREE HORIZONTAL FIELD MAGNET SYSTEM**

MODEL: **SPLIT COIL ACCESS**

FOR: **JEFFERSON SCIENCE ASSOCIATES - JLAB**

TEST DATE: **6/17/2015**

Rated Central Field¹----- 50 kG
Rated Current ----- 93.28 Amp
Field to Current Ratio ----- 0.536 kG/Amp
Homogeneity over 1 cm DSV ----- $\pm 1.0\%$
Measured Inductance ----- 39.8 Henrys
Ramp Rate from 0 to 85A ----- 0.0254 Amps/sec (1.0 V)
Ramp Rate from 85A to 93.28A ----- 0.0031 Amps/sec (0.13 V)
Room Temperature Vertical Radial Access Port ----- 2.1" x 3.3"
Room Temperature Horizontal Axial Bore ----- 3.0"
System Cooldown Time from 300K to 4K²----- ~30 hours
Total Magnet/Switch Resistance³----- 20 Ohms
Persistent Switch Heater Current ----- 20 mA

1. Magnet not warranted for operation above 50 kG.
2. Compressor input power: 460-480 VAC/60 Hz (3 Phase)
3. All resistance measurements made at room temperature at magnet power connectors.